

Alberto Marrone

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SUMMARY

Electronics Engineering Master's student at Politecnico di Milano focused on hardware engineering, PCB design, signal integrity, and power electronics. Designed and validated a 6-layer, 90 Ω controlled-impedance USB 3.2 hub PCB from schematic through fabrication and bring-up, confirming SuperSpeed link performance on measured hardware. Also achieved simulated results of 97.6% peak efficiency and 6 $\times$ switching loss reduction on a 400W, 240 kHz GaN active clamp flyback. Hands-on experience taking boards from schematic design and PCB layout through fabrication, bring-up, and hardware validation.

EXPERIENCE

Mar 2025 – Jan 2026

Formula Student EV Team (Dynamis PRC), Politecnico di Milano

Completed PCB design workflow training using Cadence Allegro/OrCAD and gained familiarity with STM32-based embedded platforms within an automotive student engineering team.

Sept 2024 – Feb 2025

Thesis Project, University of Bologna

Led end-to-end hardware development of a 240 kHz GaN-based active clamp flyback microinverter (400W). Reduced primary switching losses 6 $\times$ (from 0.12W to 0.02W) through device-level optimization; achieved 97.6% peak efficiency at 160W and 92.3% at 400W via magnetic modeling (ETD34) and high-frequency layout refinement. Designed 4-layer PCB in KiCad: minimized power loops, EMI-aware routing, controlled return paths, and isolated sensing (AMC0311S). Refined manufacturer SPICE models for improved transient accuracy.

Feb 2024 – Jun 2024

Research Intern – Power Electronics, University of Bologna

Designed and analytically modeled a 400W resonant LLC converter, including full resonant tank analysis and system-level loss evaluation. Built a reusable C/MATLAB sizing tool (Q-m gain curves, ZVS/ZCS boundary analysis) to eliminate manual calculation error across design iterations. Evaluated peak efficiency up to 97.9% across a 100 kHz–1 MHz operating range.

SELECTED TECHNICAL PROJECTS

Mar 2026 – Jun 2026

USB 3.2 Gen 1 4-Port Hub, Independent Project

Designed a fully bus-powered USB 3.2 Gen 1 (5 Gbps) hub around the TI TUSB8044A featuring hardware-only USB-C attach detection, cold-socket VBUS control, and per-port current limiting without firmware. Engineered a 6-layer impedance-controlled PCB with 90 Ω differential routing and ≤ 0.15 mm intra-pair skew; manufacturer DFM review caught and corrected a via-in-pad issue pre-fabrication. Validated the fabricated board end-to-end: all power rails within 1% of target, both USB interfaces enumerating on first power-up, and 102 MB/s SuperSpeed throughput confirmed via functional testing. [GitHub Repository](#)

Apr 2026 – Present

DRSSTC Fiber-Optic Interrupter, Independent Project

Hand-soldered every component on a galvanically isolated interrupter control board (STM32F405) for a Tesla coil driver, with a dual-redundant hardware safety chain: a PMOS cutoff de-energizes the fiber-optic output independent of firmware state, backed by a hardware interrupt halting PWM timers. Routed a 4-layer controlled-impedance stackup (90 Ω USB, 50 Ω FSMC/SDIO). Verified power rails, MCU programming, and USB communication during initial bring-up, then diagnosed and repaired PCB trace damage caused while desoldering a connector; finishing final rework before completing full bring-up. [GitHub Repository](#)

EDUCATION

Feb 2025 – Present

Politecnico di Milano, Master's Degree in Electronics Engineering

Relevant Coursework: Digital Integrated Circuit Design, Analog Circuit Design, Hardware Architectures for Embedded and Edge AI, Design of Hardware Accelerators.

Feb 2026 – Jun 2026

Universitat Politècnica de València, Erasmus Exchange Program (Semester 2 of PoliMi Master's)

Relevant Coursework: Embedded Systems for IoT, Electric Motors Efficiency, Electric Vehicles, PCB Design.

Sep 2021 – Feb 2025

University of Bologna, Bachelor's Degree in Electronics Engineering

Relevant Coursework: Power Electronics, Signal Processing, Electronic Measurements, Computer Architectures.

SKILLS

Hardware & Power Systems: Board-level power architecture, GaN/SiC converters, DC-DC/DC-AC topologies, resonant converters, PDN design, feedback control, loss modeling, efficiency optimization.

PCB Design: KiCad, Altium Designer, Cadence Allegro/OrCAD; multilayer HDI routing, high-speed digital routing, impedance control, return path control, EMI-aware layout.

Signal & Power Integrity: Parasitic modeling, switching transient analysis, snubber optimization, SPICE-to-hardware correlation, Saturn PCB Toolkit.

Simulation: LTspice, MATLAB, COMSOL Multiphysics, CST Studio Suite, AWR, PSpice.

Programming & Tools: C, Python, MATLAB scripting, Git/GitHub.

Laboratory: Oscilloscopes, spectrum analyzers, nanoVNA, programmable power supplies, electronic loads.

CERTIFICATIONS

Cadence Student Ambassador | Cadence Allegro X Advanced Package Designer | Altium Education Global Scholar | Altium PCB Design Essentials Accredited

LANGUAGES

Italian – Native | **English** – Full Professional | **Spanish** – Basic